MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY TYPE 5AMP1A

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of Specificiation MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus

See figure 1
Mounting position: any

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Rg	Zd	Ehk
Unit:	V dc	V dc	v	V dc	V dc	Meg	Meg	V dc
Maximum:	6.9	0	1,320	1,650	6,600	1.5	1	±180
Minimum:	5.7	-200	---	---	---	---	---	---
TEST CONDITIONS:	6.3	Adjust	---	Focus	2,500	---	---	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification							
5111	Vibration		---	---	Width	---	1	mm
1141	Pressure (implosion)		---	---	---	---	---	---
5216	Cathode illumination		---	---	---	---	---	---
1331	Direct-interelectrode capacitance	g1 to all	---	---	Cg1	---	5.5	pF
		k to all	---	---	Ck	---	4.9	pF
		D1 to D2	---	---	C1D2	---	3.9	pF
		D3 to D4	---	---	C3D4	---	2.5	pF
		D1 to all	---	---	CD1	---	7.9	pF
		D2 to all	---	---	CD2	---	7.4	pF
		D3 to all	---	---	CD3	---	5.5	pF
		D4 to all	---	---	CD4	---	4.8	pF
5248	Deflection-factor uniformity		---	---	---	---	1	%
5101	Neck and bulb alignement (electrostatic types)		---	---	Dia	---	2.25	in.
	Quality conformance inspection, part 1							
5201	Voltage breakdown		0.65	II	---	---	---	---
5201	Voltage breakdown (electrostatic types)		0.65	II	---	---	---	---
5206	Gas "cross" (electrostatic deflection)	Light = 15 ftL ($51.4\ cd/m^2$), See note 1	0.65	II	---	---	---	---
5106	Screen and faceplate blemishes		0.65	II	---	---	---	---
5221	Light output	Eb = 2500 V	0.65	II	---	15	---	ftL
5223	Modulation	Light = 15 ftL ($51.4\ cd/m^2$)	0.65	II	ΔEc	---	45	V dc
5231	Spot position (electrostatic deflection)		0.65	II	---	---	16	mm
5231	Spot displacement (leakage)		0.65	II	Displ	---	10	mm
5241	Grid-cutoff voltage		0.65	II	Eco	-34	-56	V dc
5251	Grid No. 1 leakage current		0.65	II	---	---	---	---
5251	Anode No. 2 leakage current		0.65	II	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 -Continued							
---	Pattern distortion	See note 2	0.65	II	---	---	---	---
---	Centering of scan	Focused trace; Ec1 at 10 V drive (see note 3)						---
		1D2	0.65	II	---	±2	---	in.
		3D4	0.65	II	---	±1.25	---	in.
	Quality conformance inspection, part 2							
1301	Heater current		---	---	If	540	660	mA
5201	Electrode current (anode No. 1)	Ec1=0	---	---	Ib1	-15	15	µA dc
5201	Electrode current (anode No. 2)	Light = 15 ftL (51.4 cd/m^2)	---	---	Ib2	---	800	µA dc
5201	Electrode current (cathode)	Light = 15 ftL (51.4 cd/m^2)	---	---	Ik	---	1,000	µA dc
5101	Base alinement (electrostatic types)	+1D2, pin No. 5	---	---	---	---	---	---
5101	Angle between traces		---	---	---	89.2	90.8	Degrees
5101	Neck and base alinement (electrostatic types)		---	---	---	---	---	---
5216	Stray light emission (conventional types)	Eb2 = 6,600 V	---	---	---	---	---	---
5226	Line width "A" (electrostatic deflection)	Light = 15 ftL (51.4 cd/m^2)	---	---	Width	---	0.8	mm
5226	Line width "B" (electrostatic deflection)	Light = 15 ftL (51.4 cd/m^2)	---	---	Width	---	0.85	mm
5246	Focusing voltage at cutoff		---	---	Eb1	---	300	V dc
5246	Focusing voltage (zero-bias)		---	---	Eb1	0	---	V dc
5248	Deflection factor	1D2	---	---	DF	40	50	V dc/in.
5248	Deflection factor	3D4	---	---	DF	20	25	V dc/in.
5251	Heater-cathode leakage current		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3							
---	Life-test provisions	Group C Eb2 = 2,500 V Light = 15 ftL (51.4 cd/m^2)	---	---	t	500	---	hrs.
---	Life-test end points	Same conditions as initial tests						
		Line width "A"	---	---	Width	---	0.80	mm
		Line width "B"	---	---	Width	---	0.85	mm
		Modulation	---	---	ΔEc	---	45	V dc

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. All portions of a raster pattern, adjusted so its widest points just touch the sides of a 2.5 by 4.0-inch (63.5 by 101.6 mm) rectangle, will fall within the area bounded by the 2.5 by 4.0-inch (63.5 by 101.5 mm) rectangle and an inscribed 2.420 by 3.912-inch (61.47 by 99.36 mm) rectangle.
3. The useful scan in the 1D2 and 3D4 directions is measured from the tube face center.
4. Revision letters are not used to denote changes due to the extensiveness of the changes.

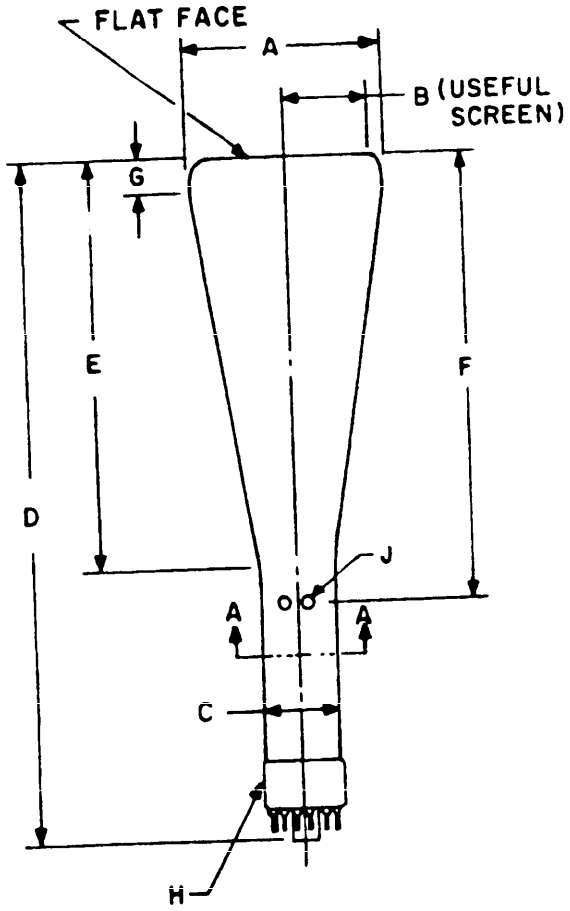
User activities:
Navy - AS, OS, MC, CG, SH

Agent:
DLA - ES

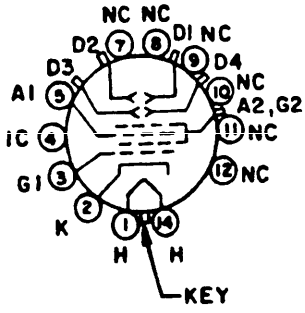
Preparing activity:
Navy - EC

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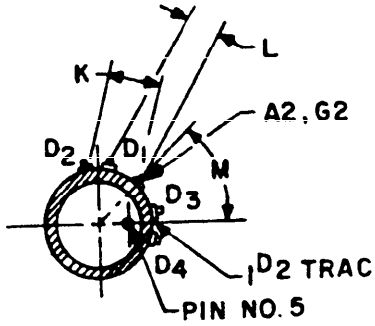


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	5.156 (130.96)	5.344 (135.74)
B	2.250 (57.15)	
C	1.938 (49.23)	2.062 (52.37)
D	17.375 (441.33)	17.875 (454.03)
E		9.0 (228.6)
F	12.412 (315.26)	12.688 (322.28)
K	1.060 (26.92)	1.180 (29.97)
L	0.500 (12.70)	0.560 (14.22)
M	40°	50°
Quality conformance inspection, part 3 (periodic check)		
H	Base: B12-37 (EIA)	
J	Caps: J1-25 (EIA)	
	Reference dimensions (see note)	
G	0.875 (22.23)	



14U

BOTTOM VIEW OF BASE



SECTION A-A

NOTE: Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 5AMP1A.

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